

Ixeia Sánchez Périz

DATA & ML ENGINEER • ML SYSTEMS • CLOUD DATA PLATFORMS

Zürich (Switzerland) • 📞 +41 78 230 9596 • ✉️ Mail

📁 Portfolio | 🔗 LinkedIn | 📄 GitHub | 🎓 Scholar

EXPERIENCE

SWISS INTERNATIONAL AIRLINES

Jun. 2024 - present | Zürich

Cloud Data Engineer • Top Performer on 1st year

- Productionized **terabyte-scale PySpark** pipelines, transforming 20k+ LoC SQL into a metadata-driven framework.
- Built an **agentic system** (code translation, validation, documentation) for on-prem to cloud migration, reducing migration time by 50% and driving AI adoption across engineering and business teams.
- Reduced deployment time by 97% (3h → 5 min), **saving 44 hours weekly** for 15 engineers through CI/CD design in Microsoft Fabric and Power BI.
- **Communication:** Presented our Data Platform Solution to the Fabric product team at **Microsoft HQ** in Redmond, US.

VISIUM (CONSULTING)

Mar. 2023 - Aug. 2023 | Zürich

Junior Machine Learning Engineer

- **Market Data:** Developed **energy hedging** method with 35% savings and constant CVaR applying forecasting & interpolation in Databricks.

CORIFY CARE (MEDTECH)

Jun. 2020 - Jul. 2021 | Valencia (Spain)

Data Scientist Intern (Part-Time 50%)

- **Signal Processing:** real-time **ML pipeline** on medical devices (64 electrodes, tested on >20 patients), achieving low-latency **inference** for ECG classification.

PROJECTS

- **Fitness AI Assistant:** Built a gym training assistant using multi-modal embeddings and stored on a vector database for semantic exercise search. MLflow for experiment tracking.

EDUCATION

ETH ZÜRICH

Sep. 2021 - Apr. 2024 | Zürich

MSc in Data Science

- **Thesis (Honours):** **Sailor**, Scalable **open source** system for LLM training in the cloud.
 - Developed **distributed training** pipelines for **1.5B+** parameter LLMs using PyTorch & DeepSpeed, on top of a resource planner. Built on top of Kubernetes (GKE), Docker, NCCL, Gloo.
 - Implemented a novel in-CPU checkpointing mechanism (**2.87×** higher **goodput** on preemptible VMs) with parallelism.
- Cloud Computing: Designed dynamic Kubernetes **scheduler** achieving SLO 1.5ms 95th percentile latency.
- **Computer Vision:** Domain adaptation for tree canopies segmentation from drone images to **satellite imagery**.
- **Audio Data:** Automated an **ablation pipeline** to validate +250 Kaggle solutions on **self-designed** strategies.

POLYTECHNIC UNIVERSITY OF VALENCIA (UPV)

Sep. 2016 - Jul. 2021 | Valencia (Spain)

Double BS in Telecommunication Engineering & Business Administration and Management

- **Ranked 1st** (avg. 9.1/10, 33 courses with honours) • Santander Bank Scholarship

SKILLS

PROGRAMMING & FOUNDATIONS	Python, SQL (advanced), Bash, Git • Data structures, parallelization, testing (pytest)
DATA ENG & ORCHESTRATION	PySpark, Databricks, Microsoft Fabric, dbt, BigQuery, Fivetran, Milvus Vector DB CI/CD (Azure Pipelines, Terraform, GitHub Actions) • Workflow orchestration (ADF)
ML & AI	PyTorch, DeepSpeed, Multiprocessing • LLM training, MCP, MLOps, mlflow, Gradio transformers, embeddings, agentic systems, multi-modal AI
CLOUD & INFRA	Azure, GCP (GKE), Docker, Kubernetes • Distributed computing (NCCL, Gloo)

HONORS & RECOGNITION

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| • Commencement Speaker for BSc graduation ceremony at University | May 2025 • Valencia (Spain) |
| • Youngest Speaker at a University Career guidance talk at Uni Zurich | Nov. 2024 • Zurich |
| • Best Student Award by Social Council (500+ students) | Jul. 2022 • Valencia (Spain) |
| • Nova Top 111 University talent in Spain (under 25) | Mar. 2022 • Madrid (Spain) |
| • Outstanding Graduate Award (1st in Class) | Jul. 2021 • Valencia (Spain) |

LANGUAGES

English (C1) • German (B1/B2) • French (A1) • Spanish (Native)